

The linear polynomial defeats the constant bound

Disproof of Conjecture 00000000010

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Scope of the disproof. The conjecture concerns a square-free integer polynomial f and asserts, among other clauses, the explicit upper bound

$$C_f \leq 2 \log D_f,$$

where C_f is its Bateman–Horn local-density product and D_f is the discriminant (or its absolute value). Neither language imposes a degree lower bound, excludes discriminant one, or replaces the discriminant by a truncated quantity such as $\max\{2, |\text{disc}(f)|\}$. We disprove this upper bound alone; the remaining asymptotic and optimality clauses need not be decided. In particular, the disproof uses no convention about dividing by zero in the separate lower bound.

An admissible polynomial and its discriminant. Take the actual polynomial $f(X) = X \in \mathbb{Z}[X]$. It has degree one, is primitive, has positive leading coefficient and has no repeated root. In particular it is square-free. The standard root-product definition of the discriminant of a split degree- d polynomial with leading coefficient a_d is

$$\text{disc}(f) = a_d^{2d-2} \prod_{1 \leq i < j \leq d} (r_i - r_j)^2.$$

Here $d = 1$, $a_d = 1$, and the single root is $r_1 = 0$. The product over root pairs is empty, so it equals one. Consequently $\text{disc}(X) = 1$ and $D_X = 1$, for either signed or absolute discriminant. This value is computed from the definition, not assigned as an auxiliary certificate.

The genuine local-density product. For each prime p , the reduction of X in $\mathbb{F}_p[X]$ has exactly one root, namely zero. Thus its root count $\rho_X(p)$ is one, and the single-polynomial Bateman–Horn factor is

$$\frac{1 - \rho_X(p)/p}{1 - 1/p} = \frac{1 - 1/p}{1 - 1/p} = 1.$$

The denominator is positive since $p \geq 2$. Every actual prime partial product

$$\prod_{p \leq N} \frac{1 - \rho_X(p)/p}{1 - 1/p}$$

therefore equals one. These partial products converge to one, proving $C_X = 1$ directly from the stated local-density product. There is no local obstruction and no unproved prime-counting assertion in this calculation.

Contradiction. The asserted upper bound at this admissible polynomial becomes

$$1 = C_X \leq 2 \log D_X = 2 \log 1 = 0,$$

which is false. Hence the conjecture, with its stated scope, is false. Adding a degree or discriminant restriction would change that scope; the present counterexample does not purport to decide such a modified claim.

Formal verification. The Lean project constructs the polynomial X over $\mathbb{Z}$, proves its square-freeness and degree, maps it into $\mathbb{C}[X]$, verifies its one-root factorization and computes the standard discriminant product. It counts the actual zeros of its reduction in $\mathbf{ZMod}\ p$ for every prime, computes each local factor, forms the prime partial products, proves their limit, and defines the constant by that limit. Its final theorem disproves the stated logarithmic upper bound.